Multipole Expansion: Five Charge Distributions

Student Workbook — Guided Mathematical Development

Name: _____ Date: _____

What you will learn

Recognize source geometry, choose the correct charge element, use symmetry to predict vanishing moments, evaluate the remaining moments, and interpret the far-field potential. Work with the Lesson 14 review guide open.

Coordinate and notation reminder

As in the lecture, θ is the polar angle from $+z$ and ϕ is azimuth. Your preferred convention reverses these symbols. A prime marks source coordinates, not a derivative.

Useful Legendre values

$P_0(x) = 1$, $P_1(x) = x$, $P_2(x) = (3x^2 - 1)/2$, $P_3(x) = (5x^3 - 3x)/2$, $P_4(x) = (35x^4 - 30x^2 + 3)/8$. Also $P_n(1) = 1$ and $P_n(-1) = (-1)^n$.

Activity 1: Choose the correct multipole setup

Where this fits

Every example begins with source geometry, the charge element, and the convergence region. The index n labels angular patterns, not individual charges.

1a.

Explain the meaning of r, r', θ', dQ, Q_n, and $R_{\max}$. Write the moment formula and the first three terms of the potential.

Guided starting point

Distinguish a small piece of physical charge dQ from its weighted moment Q_n.

1b.

Match each source to its charge element: point charges, a line, a ring, and a spherical cloud. Explain how to check normalization.

Guided starting point

For a ring $d\ell' = a\,d\phi'$; for a cloud $d\tau' = r'^2 \sin\theta' dr' d\theta' d\phi'$.

1c.

Predict which moments vanish for a distribution unchanged under $z \mapsto -z$, and for one whose signed charge reverses under that reflection. Justify using $P_n(-x) = (-1)^n P_n(x)$.

Guided starting point

Pair source elements at opposite z with the same r'.

Activity 2: A single point charge at the origin

Where this fits

Begin with the simplest source. This example also reveals why the multipole moments depend on the chosen origin.

2a.

For a point charge $+q$ at the origin, identify $R_{\max}$ and evaluate Q_0 without a volume integral.

Guided starting point

Use the discrete-charge sum and $P_0 = 1$.

2b.

Explain why every Q_n with $n \geq 1$ vanishes, even though the polar angle at the origin is undefined.

Guided starting point

The factor $r'^n P_n(\cos\theta')$ tends to zero as $r' \to 0$ for $n \geq 1$.

2c.

Write the complete exterior potential and check its angular dependence and domain. What changes if the coordinate origin moves?

Guided starting point

Only Q_0 survives for the centered charge.

Activity 3: Two opposite charges on the polar axis

Where this fits

For the axial dipole, the charge signs and position signs both matter. Calculate the first moments explicitly before generalizing.

3a.

Draw $+q$ at $z = d/2$ and $-q$ at $z = -d/2$. Label their radii and polar angles, then determine $R_{\max}$.

Guided starting point

Both source radii equal $d/2$, but their polar angles are 0 and π.

3b.

Calculate Q_0, Q_1, Q_2, and Q_3 separately, showing the contribution from each charge. Explain the parity pattern.

Guided starting point

Use $P_n(1) = 1$ and $P_n(-1) = (-1)^n$.

3c.

Write the first two nonzero potential terms. Identify the leading far-field term and explain its direction.

Guided starting point

The dipole moment points from the negative charge to the positive charge.

Activity 4: A spherical charge cloud

Where this fits

The density $f(r')$ can vary radially. Spherical symmetry, not uniform density, removes the higher exterior moments.

4a.

For $\rho = f(r')$ inside $r' \leq a$, write dQ, the integration limits, and the normalization equation for total charge q.

Guided starting point

The volume element contains $r'^2 \sin\theta'$.

4b.

Factor the moment integral into radial and angular pieces. With $u = \cos\theta'$, show why $Q_n = 0$ for $n \geq 1$.

Guided starting point

Show the transformed limits $\theta' = 0 \to u = 1$ and $\theta' = \pi \to u = -1$.

4c.

Give the exterior potential and explain why the same expression is not automatically valid inside the cloud.

Guided starting point

The exterior series requires $r > a$; interior charge contributes differently.

Activity 5: A uniformly charged circular ring

Where this fits

All ring elements have the same radius and polar angle. Geometry makes the moment integral simple; the angular sign is important.

5a.

Sketch the ring in the xy plane. Label its radius a, polar angle θ', line density λ, and charge element dQ. Verify the total charge.

Guided starting point

Use $\theta' = \pi/2$, $d\ell' = a\,d\phi'$, and $\lambda = q/(2\pi a)$.

5b.

Derive $Q_n = qa^n P_n(0)$. Compute Q_0, Q_2, and Q_4 and explain why odd moments vanish.

Guided starting point

Use $P_2(0) = -1/2$ and $P_4(0) = 3/8$.

5c.

Write the first three nonzero terms of $V(r,\theta)$. Check the axis result against the exact ring potential $V(r,0) = q/(4\pi\epsilon_0\sqrt{r^2+a^2})$.

Guided starting point

Factor out $1/r$ and expand $(1+a^2/r^2)^{-1/2}$.

Activity 6: A finite uniformly charged axial line

Where this fits

Unlike the ring, the line lies on the polar axis. Its quadrupole has the opposite sign even though both carry positive charge.

6a.

Sketch the line from $z=-a$ to $z=a$. Find λ, $R_{\max}$, and the two polar angles needed to split the integral.

Guided starting point

Upper half: $\theta' = 0$; lower half: $\theta' = \pi$.

6b.

Split the moment into upper and lower integrals over $0 \le r' \le a$. Integrate and obtain the general even/odd result.

Guided starting point

Use $\int_0^a r'^n dr' = a^{n+1}/(n+1)$.

6c.

Find Q_0, Q_2, and Q_4, write the exterior series through $n = 4$, and compare the sign of Q_2 with the ring.

Guided starting point

On the polar axis $P_2(\pm 1) = 1$, whereas the ring has $P_2(0) = -1/2$.

Activity 7: Compare, combine, and choose the leading term

Where this fits

Use symmetry and superposition to solve a new problem without starting the integration over again.

7a.

Complete a comparison: for the five lesson sources, give $R_{\max}$, which moments survive, and the leading far-field potential.

Guided starting point

For nonzero net charge, the monopole is leading even when higher moments are present.

7b.

A concentric spherical cloud carries $+q$ and a uniformly charged ring of radius a carries $-q$. Find Q_0 and Q_2 and the leading far-field potential.

Guided starting point

The cloud has only a monopole; the ring has $Q_2 = q_{\text{ring}} a^2 P_2(0)$.

7c.

For a generic source, state three checks that can catch an incorrect moment calculation: normalization, units, and convergence domain. Explain how each check works.

Guided starting point

$[Q_n] = \text{C}\,\text{m}^n$ and the series needs $r > R_{\max}$.